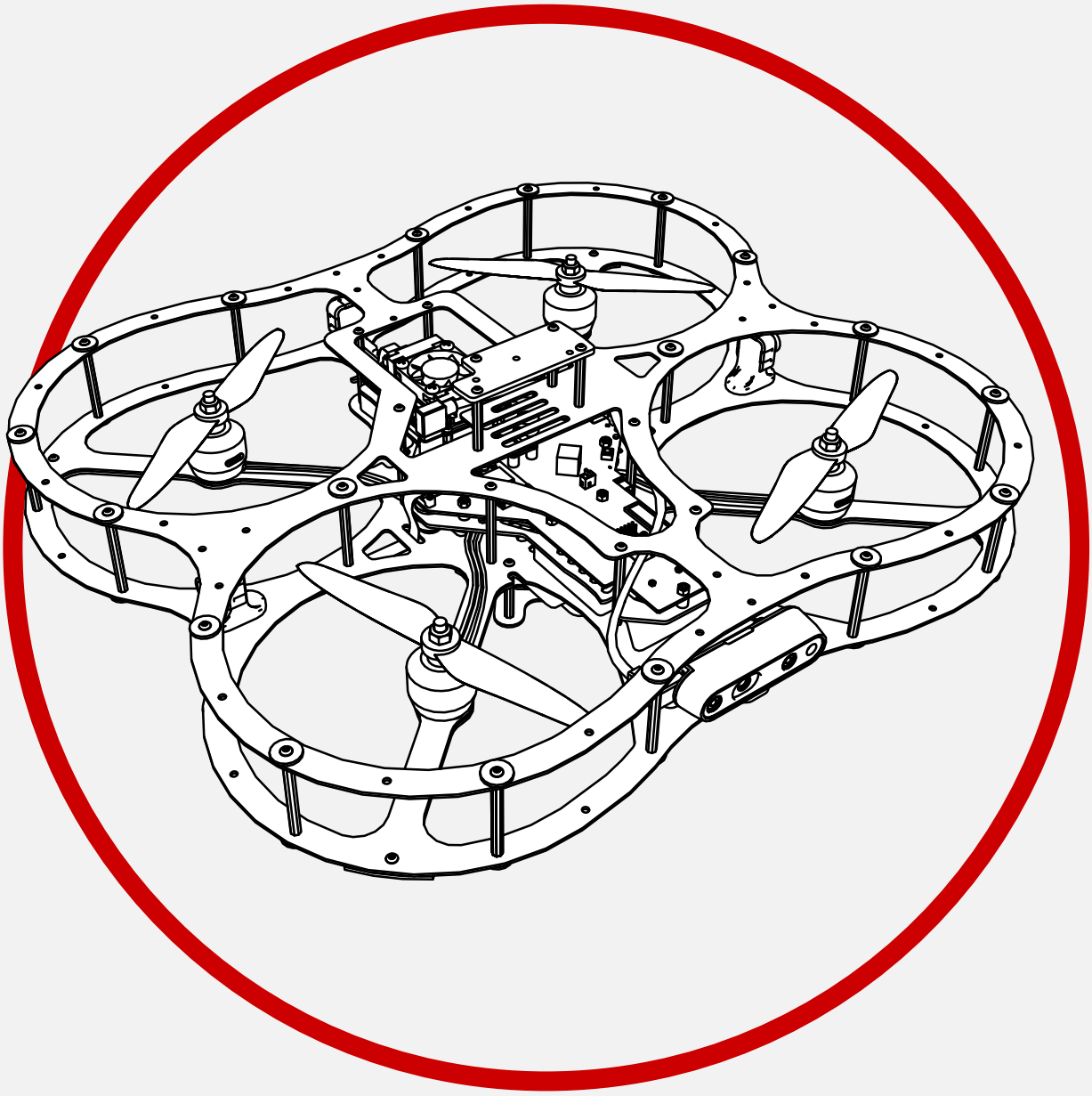

Autonomous Vehicles Research Studio

Setup Guide – QDrone 2 Communication

$v\ 2.0\ -\ 18th\ April\ 2023$

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. **NOTE:** While the GPIO, and USB ports provides connections for external user devices, users are responsible for certifying any modifications or additions they make to the default configuration.

FCC Notice This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.
Contains FCC ID: SQG-60SIPT

Industry Canada Notice This Class A digital apparatus complies with CAN ICES-3 (A). Cet appareil numérique de la classe A est conforme à la norme NMB-3 (A) du Canada.
Contains IC: ST60-2230C-PU

Waste Electrical and Electronic Equipment (WEEE)

This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

CE Compliance

This product meets the essential requirements of applicable European Directives as follows:

- 2014/35/EU; Low-Voltage Directive (safety)
- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)
- 2014/53/EU; Radio Equipment Directive (RED)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Caution

During flight QDrone 2 sound pressure level has been measured at 92 dBA at 1m away from the QDrone 2 and it is considered hazardous. Users shall ensure that they are not exposed to a sound level greater than the hazardous level as defined by the local authority. Use protective earpieces during operation.

Caution

The Intel RealSense D435 RGB-D camera is classified as a Class 1 Laser Product under the IEC 60825-1, Edition 3 (2014) internationally and EN 60825-1:2014+A11:2021 in Europe. The camera complies with FDA performance standards for laser products except for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019.
Do not power on the product if any external damage is observed. Do not open or modify any portion of any laser product as it may cause the emissions to exceed Class 1. Invisible laser radiation when opened. Do not look directly at the transmitting laser through optical instruments such as a magnifying glass or microscope. Do not update laser product firmware unless instructed by Quanser.

Regular maintenance of QDrone 2:

- Inspect the propellers before flight to confirm they are not damaged or loose (able to move while the motor is not moving).
- Prior to using the QDrone 2, visually inspect the LiPo battery for damage (e.g., bloating). **DO NOT USE** the battery if damaged.
- Ensure that the battery and its cables are secured using the provided straps to avoid movement or damage during flight.
- Inspect the QDrone 2 frame before and after each flight to confirm that no major structural damage exists. Repair if needed.

Table of Contents

A. Communicating with QDrone 2 3

B. Boot-Up for QDrone 2 4

i. Connecting the Battery 4

ii. Turning the QDrone 2 ON and OFF 4

iii. Testing the Connection 5

A. Communicating with QDrone 2

The QDrone 2 is shipped pre-configured to connect to the wireless network created by the provided router: Quanser_UVS-5G. This happens automatically following a boot sequence when they are powered on. To ensure that the drone is connected, observe if there is an IP in the LCD screen on top of the drone and try to ping it from the command prompt in the ground control station, similar to ensuring that the ground control station PC - router connection has been established in the router to PC documentation step 7 – router pc connection.

To connect additional vehicles to the UVS network, the 5GHz and 2.4GHz bands on the router have been configured as follows:

5GHz:

SSID: Quanser_UVS-5G Password: UVS_wifi

2.4GHz:

SSID: Quanser_UVS Password: UVS_wifi

Router login credentials are as follows:

Username: admin Password: Quanser_123

The QDrone 2 does not have a preset IPV4. **If you would rather set a fixed IP, refer to the Setting Fixed IP Addresses document (supplementary_material > setting_fixed_ip).** For a successful connection, the DHCP server option on the router must be enabled. For the Netgear Nighthawk router provided with the AVRS system, the DHCP server can be found by going to Advanced/Setup/LAN Setup.

To ensure compatibility with the Self-Driving Car Research studio, the **5GHz band** for the Netgear Nighthawk router has been configured to **channel 44.** If you do notice intermittent issues with communication to any of the vehicles, it is recommended that you use a WiFi spectrum analyzer and check if there are networks which are broadcasting on the same channel but at a higher signal strength. Microsoft has a free WiFi analyzer: **(** https://www.microsoft.com/en-us/p/wifi-analyzer/9nblggh33n0n?activetab=pivot:overviewtab#)

You can change the Netgear Nighthawk's channel number by logging into the router and checking the channel number under the 5GHz wireless band.

B. Boot-Up for QDrone 2

i. Connecting the Battery

Insert a fully charged battery into the battery compartment on the QDrone 2 (Figure 1) all the way to the hard stop marked with a red circle at the bottom of the image. Make sure that the long cables that have the XT-60 connector go to the top left so that you can connect it plug it in, **Tighten the battery with the velcro**. It is generally good practice to hide the 5-pin connector under the battery, so it is not dangling while in flight.

> **Note:** Ensure that the velcro strap is tight and that the battery is secured in place.

Connect the XT-60 connector on the battery to the XT-60 connector on the tail at the bottom of the drone (Figure 1).

> ⚠ **Caution:** Leaving the QDrone 2 powered on with the battery connected will continue to drain power below the minimum voltage threshold of 13.3V **and may permanently damage the battery**.

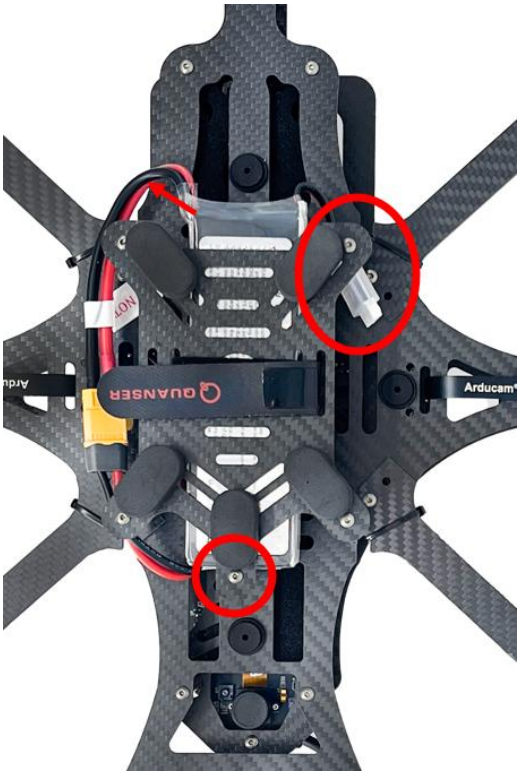


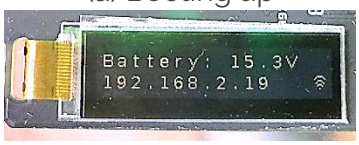
Figure 1. Battery setup

ii. Turning the QDrone 2 ON and OFF

Press and quickly release the red power button on the drone PCB to turn it on. It should be acknowledged by 3 beeps from the QDrone 2 as the drone's computer boots. The LCD on the QDrone 2 should turn on and it should show Figure 2a. Figure 2 shows the LCD display as the QDrone 2 is being turned on (2a), as it starts (2b) and gets an IP address (2c) and what happens after the button is pressed for power off (2d).

(a) Booting up

(b) Started



(c) Connected

(d) Powering off

Figure 2: QDrone 2 boot and power off

> **Note:** Turn off the QDrone 2 by using the red power button. Only press once quickly and release, LCD should show Figure 2d. Do not keep the red button pressed to turn it off as it could cause issues. Disconnect the XT-60 battery cable whenever the QDrone 2 is not in use.

iii. Testing the Connection

Open a command prompt on the ground control station PC (type `cmd` in the start menu). Type the following command: `ping 192.168.2.d -t` where 192.168.2.d represents the IP on the drone. It should be displayed in the LCD when the drone is powered on. It could take a couple of minutes to appear. A reply should be registered as in Figure 3, which indicates that a connection has been established. You can press CTRL+C to terminate the ping.

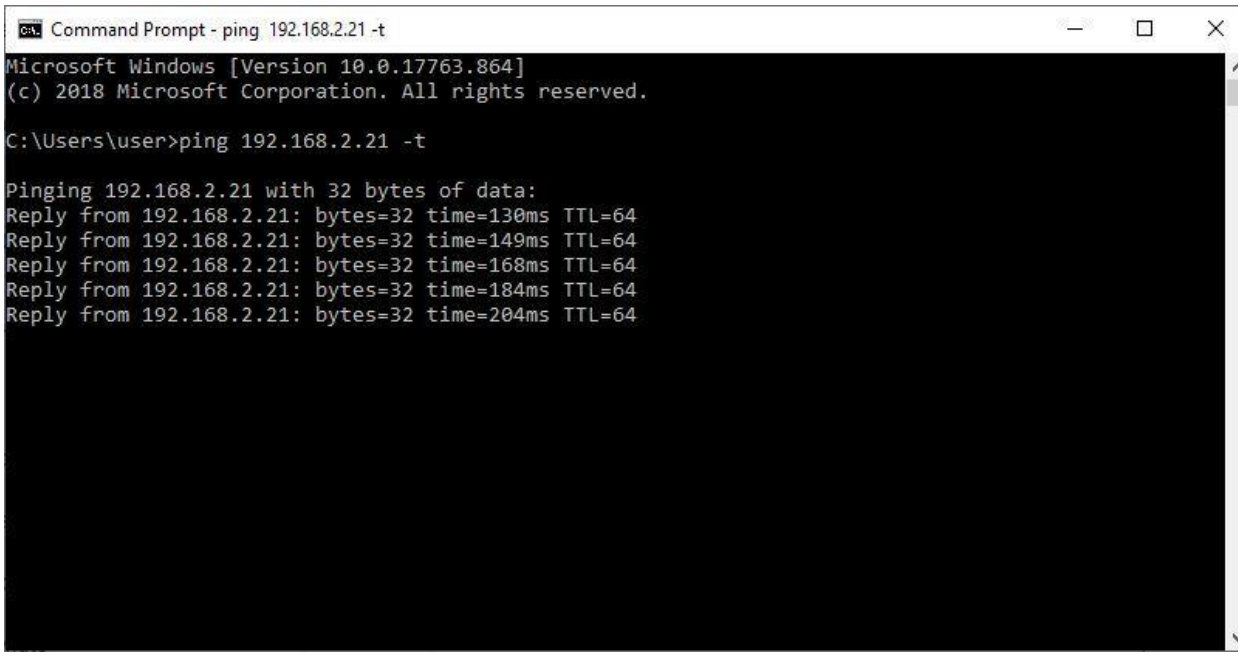


Figure 3: Checking the connection between the QDrone 2 and the ground control station PC

An alternative way of checking connectivity between the QDrone 2 and the router is to note if the router is seeing the QDrone 2 by connecting to the router. Router login credentials are in section A.

The QDrone 2 network information can be found by checking under the Connected/Attached Devices option.

If the QDrone does not have an IP address on its LCD after a few minutes and the router is turned on configured properly, contact Quanser's technical support.

> **Note:** If the ping test fails, double-check the network connection and try again. Also try power cycling the drone. If issues persist, contact Quanser technical support (tech@quanser.com).

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